（回顾）

@ 晶体结合-范德瓦尔斯结合：

（3）非极性分子之间的作用

氢键

混合键

@ 分子动力学

热容，摩尔热容

能量均分定理

单原子，双原子，固体

热膨胀

@ 热膨胀，线膨胀系数，体积膨胀系数

@ 玻尔兹曼能量分布

@ 噪声涨落（固体内原子能量应该为3kT）

$\frac{1}{2}M\overline{V^2}=\frac{3}{2}kT$ ⇒ $\frac{1}{2}MV^2=3kT$

订正

@ 阿雷尼乌斯方程，热激活能，扩散系数

$$D=D_0\exp\left(-\frac{E_A}{kT}\right)$$

@ 晶体结构

（一） 原胞与基矢(primitive cell and unit vector)

（二） 晶胞(crystal unit cell)

（三） 简单晶格与复杂晶格

(simple and complex crystal lattice)

（四） 布拉伐格子(Bravais lattice)

由一组基矢所决定的平行六面体所围起来的最小重复单元就叫原胞（或初基单胞Primitive Cell）。(固体物理常用) The smallest repeating unit enclosed by a parallelepiped determined by a set of base vectors is called a primitive cell. (Commonly used in solid-state physics)

原胞代表的是几何形状、体积的最小重复单元，不具有物理内涵，而基元则有物理内涵。 The primitive cell represents the smallest repeating unit of geometric shape and volume, without physical meaning, while the basis has physical meaning.

原胞往往不能反映晶体的对称性，习惯上常选择能反映晶体对称性的重复单元，这种重复单元就叫晶胞（conventional cell）。晶胞一般不是最小的重复单元。其体积（面积）可以是原胞的数倍。 The original cell often cannot reflect the symmetry of the crystal, and it is customary to choose repeating units that can reflect the symmetry of the crystal, which are called conventional cells. A unit cell is generally not the smallest repeating unit. Its volume (area) can be several times that of the original cell.

选取原胞的原则是 反映周期性，反映对称性，体积最小，这些原则一般 不能兼顾，所以选取方法仍不唯一。只要求体积最小就得到了原胞 原胞的选择也不是唯一的，但是必须只包含一个节点，是晶体结构的最小单元，选取原则 是尽量使得各边都是点阵的最短平移矢量。The principle of selecting primitive cells is to reflect periodicity, symmetry, and minimum volume. These principles generally cannot be taken into account simultaneously, so the selection method is still not unique. The selection of a primitive cell that only requires the smallest volume is not unique, but it must contain only one lattice, which is the smallest unit of the crystal structure. The selection principle is to make each edge the shortest translation vector of the lattice as much as possible.

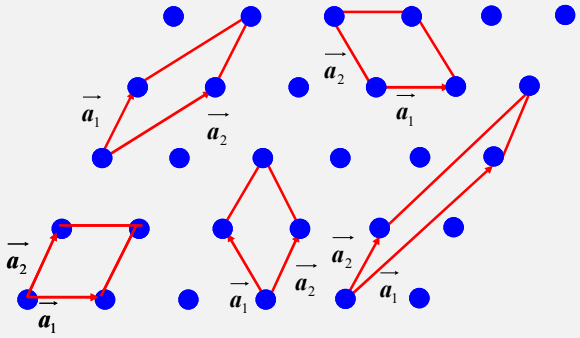
能反映晶体对称性的最小重复单元叫威格纳-赛兹原胞（Wigner-Seitz Cell）。它按以下方法选取：最近邻或次近邻两格点间连线的垂直平分面(三维)或垂直平分线(二维)所围成的原胞。The smallest repeating unit that can reflect crystal symmetry is called the Wigner Seitz cell. It selects the primitive cell enclosed by the perpendicular bisection plane (three-dimensional) or perpendicular bisector (two-dimensional) of the lines connecting the nearest or second nearest grid points.

（一）原胞与基矢(primitive cell and unit vector)

原胞:一个晶格中最小重复单元
Primitive cell: the smallest repeating unit in a lattice

基矢:原胞的边矢量
Base vector: the edge vector of the primitive cell

三维晶格的重复单元是平行六面体
The repeating unit of a three-dimensional lattice is a parallelepiped



原胞及基矢的选取不是唯一的 The selection of primitive cells and basal vectors is not unique

01/19

（二）晶胞(Unit cell)

01 定义 | Definition

晶体结构中具有代表性的最小重复单元，包含了晶体的全部结构信息。它是晶体周期性排列的基础，其形状、大小和原子位置决定了晶体的几何特征。

The smallest repeating unit in a crystal structure that contains all the structural information of the crystal. It is the basis of the periodic arrangement of crystals.

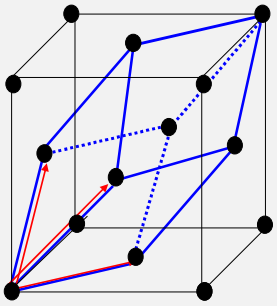
02 概念 | Concept

在三维空间中沿x、y、z轴方向无限且无间隙地重复排列晶胞，即可构建出完整的宏观晶体。它是连接晶体微观结构与宏观物理性质的关键纽带。

By repeating the unit cell infinitely and without gaps in three-dimensional space along the x, y, and z axes, the entire macroscopic crystal can be constructed.

核心总结：晶胞是晶体结构的“基石”，理解其特征是掌握晶体化学与材料科学的第一步。

例如：面心立方晶格为反映整个格子的立方对称性，选图中的立方体为其晶胞。**For example, a face centered cubic lattice reflects the cubic symmetry of the entire lattice, and the cube in the figure is selected as its unit cell.**



（三）简单晶格与复式晶格(simple and complex crystal lattice)

简单晶格 Simple lattice

由一种原子构成的单一结构晶格 only one kind of atom and one lattice

复式晶格 Complex lattice

由一种原子构成的多结构晶格 only one kind of atom but more lattices

由多种原子构成的单一结构晶格 two or more kinds of atoms in one lattice

简单晶格 Simple lattice

由完全等价的一种原子构成的晶格 only one kind of atom

I 简单立方晶格 simple cubic lattice

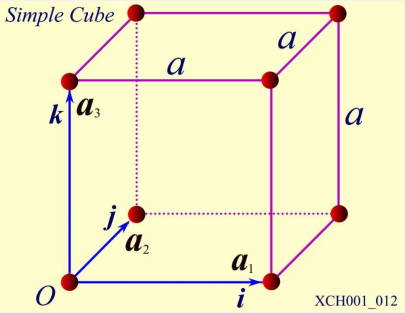
原胞为简单立方晶格的立方单元 **The primitive cell is a cubic unit with a simple cubic lattice**

$\vec{a}_1 = a\vec{\imath},\ \vec{a}_2 = a\vec{\jmath},\ \vec{a}_3 = a\vec{k}$

原胞体积 **cell valume**

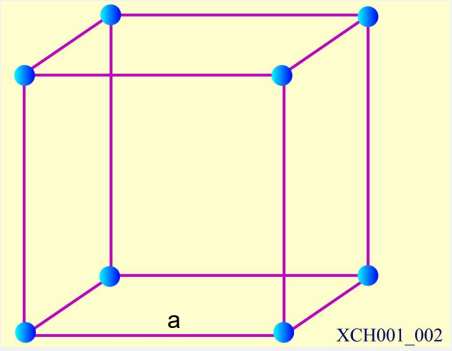
$V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3) = a^3$

原胞中只包含一个原子 **only one atom in a primitive cell**



Simple cubic 简单立方

Eight atoms at the corners. 原子球占据立方体的8个顶点；
Coordination number(Cn,配位数）=6
Lattice parameter a (晶胞参数a)



No metals

II 面心立方晶格 face centered cubic lattice

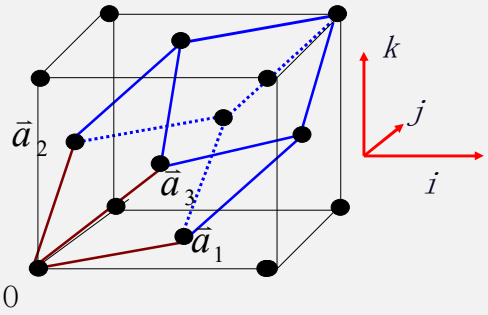
立方体的顶点到三个近邻的面心引三个基矢three base vectors from a cornor to the center of its three nearest neighbors' faces

原胞基矢 base vectors(primitive cell)

$$\vec{a}_1=\frac{a}{2}(\vec{i}+\vec{j})$$

$$\vec{a}_2=\frac{a}{2}(\vec{j}+\vec{k})$$

$$\vec{a}_3=\frac{a}{2}(\vec{k}+\vec{i})$$



原胞的体积 volume(primitive cell)

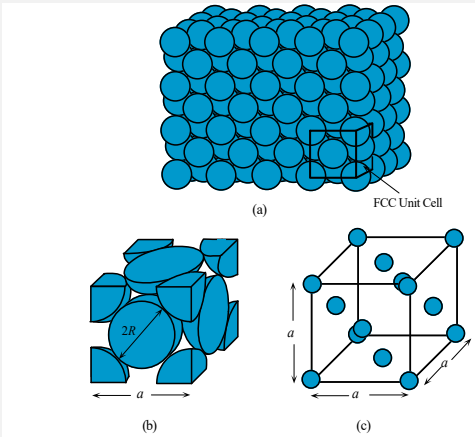
$$V=\vec{a}_1\cdot(\vec{a}_2\times\vec{a}_3)=\frac{1}{4}a^3$$

晶胞基矢 unit cell vectors

$$\vec{a}_1=a\vec{\imath},\ \vec{a}_2=a\vec{\jmath},\ \vec{a}_3=a\vec{k}$$

晶胞的体积 unit cell volume

$$V=\vec{a}\cdot(\vec{b}\times\vec{c})=a^3$$



(a) The crystal structure of copper is Face Centered Cubic (FCC). The atoms are positioned at well defined sites arranged periodically and there is a long range order in the crystal.
(b) An FCC unit cell with closed packed spheres.
(c) Reduced sphere representation of the FCC unit cell. Examples: Ag, Al, Au, Ca, Cu, γ-Fe (>912°C), Ni, Pd, Pt, Rh

Face-centered cubic (FCC). 面心立方

The unit cell of the copper crystal is cubic with Cu atoms at its **corner** and one Cu atom at the **center of each face**.

Lattice parameter a (晶胞参数 a）: the length of the cubic unit.

For Cu: a=0.362nm
R=0.128nm
$\sqrt{2}$a=4R

How many atoms in the FCC unit cell?
Coordination number?

8*1/8+6*1/2=4

Cn=12

FCC structure: close-packed crystal structure (密堆积结构）。The volume of the FCC unit cell is 74 percent full of atoms, which is the maximum packing possible with identical spheres.

Metallic - fcc

- Atomic packing factor:

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

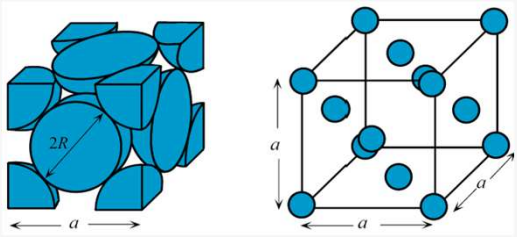
*assume hard spheres

原子堆积因子

- Calculate the APF for the FCC crystal structure

$$APF = \frac{4 \times \frac{4}{3}\pi r^3}{a^3} \approx 74\%$$

常见晶体结构：金属 - FCC



图示呈现了面心立方（FCC）结构的原子堆积模型与晶胞几何关系。该结构中原子紧密排列，不仅占据立方体的8个顶点，还分布在6个面的中心，这种排列方式赋予了金属良好的延展性与强度。

结构特征

原子分布在立方体的8个顶点和6个面心，具有极高的对称性，是金属晶体中常见的密排结构之一。

晶胞原子数

顶点贡献：8×(1/8)=1；面心贡献：6×(1/2)=3。
总计：**4个原子/晶胞**。

堆积效率

空间利用率高达74%，配位数为12（每个原子被12个等同原子包围），属于最密排结构。

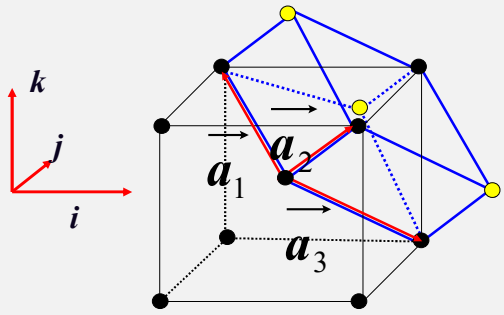
典型金属

铜(Cu)、铝(Al)、金(Au)、银(Ag)、镍(Ni)等。这类金属通常表现出优异的塑性、导电性和导热性。

III 体心立方晶格 body centered cubic

基矢：由立方体的中心到三个顶点引三个基矢
base vectors: from the center of the cube to its three vertices

$$\vec{a}_1 = \frac{a}{2}(-\vec{\imath} + \vec{\jmath} + \vec{k})$$
$$\vec{a}_2 = \frac{a}{2}(\vec{\imath} - \vec{\jmath} + \vec{k})$$
$$\vec{a}_3 = \frac{a}{2}(\vec{\imath} + \vec{\jmath} - \vec{k})$$

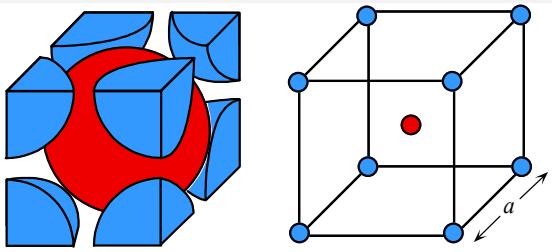


原胞体积 volume of primitive cell

$$V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3) = \frac{1}{2}a^3$$

原胞中只包含一个原子 **one atom inside**

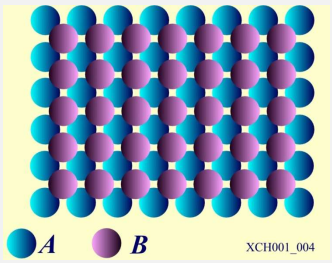
Body-centered cubic(BCC)体心立方



a　　b

Examples: Alkali metals (Li, Na, K, Rb), Cr, Mo, W, Mn, α-Fe (< 912 °C), β-Ti (> 882 °C).

Body centered cubic (BCC) crystal structure. (a) A BCC unit cell with closely packed hard spheres representing the Fe atoms. (b) A reduced-sphere unit cell.



a=?R

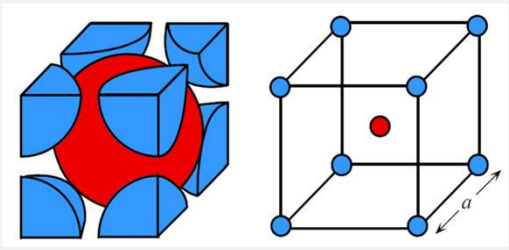
Cn=?

$a = \frac{4R}{\sqrt{3}}$；　Cn=8；

How many atoms in a unit cell?　2

one atom at the center of the cell, eight atoms at the corners.
The volume of the BCC unit cel[68]?　percent full of atoms.

常见晶体结构：金属 - BCC



体心立方（BCC）是金属晶体中常见的结构形式。其核心特征是除了立方体的八个顶点各有一个原子外，立方体的几何中心还存在一个额外的原子，这种排布方式赋予了金属独特的物理与力学性能，如较高的熔点和良好的塑性。

结构排布特征

原子占据立方体的8个顶点和1个体心位置，具有体心立方的空间对称特性，晶胞形状为正六面体。

晶胞原子数 (Z=2)

顶点原子贡献：8×1/8=1，体心原子贡献：1×1=1，总计每个晶胞含2个完整原子。

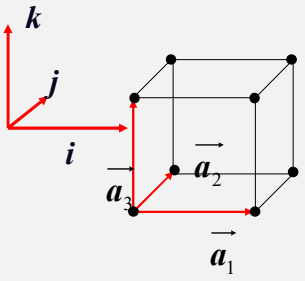
堆积密度 (68%)

原子总体积占晶胞体积的68%，属于非密排结构，致密度低于面心立方(FCC)，原子间空隙较大。

典型金属实例

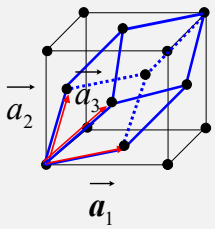
常见于过渡金属，如铁(Fe)、铬(Cr)、钨(W)、钼(Mo)、钒(V)等，这类金属通常具有较高的熔点和硬度。

核心应用：BCC结构的金属因原子排列较为松散，原子扩散速率较快，在高温环境下仍能保持较好的强度，常被用作耐高温结构材料及合金基体。



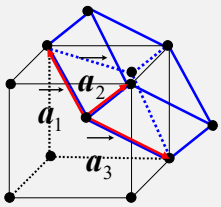
SC简立方结构原胞

$$\vec{a}_1 = a\vec{i}$$
$$\vec{a}_2 = a\vec{j}$$
$$\vec{a}_3 = a\vec{k}$$



FCC面心立方原胞

$$\vec{a}_1 = \frac{a}{2}(\vec{i} + \vec{j})$$
$$\vec{a}_2 = \frac{a}{2}(\vec{j} + \vec{k})$$
$$\vec{a}_3 = \frac{a}{2}(\vec{k} + \vec{i})$$

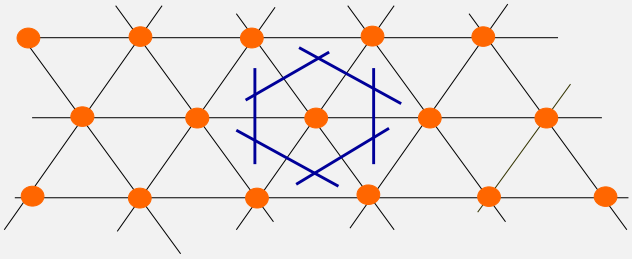


BCC体心立方原胞

$$\vec{a}_1 = a/2(-\vec{i} + \vec{j} + \vec{k})$$
$$\vec{a}_2 = a/2(\vec{i} - \vec{j} + \vec{k})$$
$$\vec{a}_3 = a/2(\vec{i} + \vec{j} - \vec{k})$$

维格纳—赛兹原胞

以某一格点为中心，作它与最近邻、次近邻等格点的垂直平分面，由这些面所围成的封闭多面体称维格纳—赛兹原胞，也满足原胞的要求，而且每个维格纳—赛兹原胞只含有一个格点并位于原胞的中心。 A closed polyhedron formed by dividing a lattice point into perpendicular bisectors with its nearest and second nearest neighbors is called a Wigner Seitz primitive cell, which also meets the requirements of a primitive cell. Each Wigner Seitz primitive cell contains only one lattice point located at the center of the primitive cell.



三种原胞特点properties of three kinds of cells

固体物理学原胞 Primitive cell

是体积最小的原胞，格点只在顶角上，每个原胞平均只含一个格点，其基矢常用 $\vec{a}_1,\vec{a}_2,\vec{a}_3$ 来表示；The smallest primitive cell in terms of volume, with lattice points only located at the top corners. Each primitive cell contains an average of only one lattice point, and its basis vector is written as $\vec{a}_1,\vec{a}_2,\vec{a}_3$

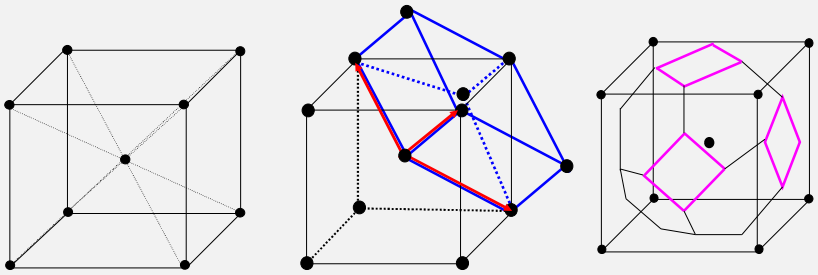
结晶学原胞 Unit cell

为同时反映晶体的微观周期结构和晶体对称性，通常选取体积较大的原胞，常称为“晶胞”，其格点不只在顶角上，还可分布在体心和面心处，每个原胞平均不只含一个格点，其基矢常用 $\vec{a},\vec{b},\vec{c}$ 来表示；Reflect both the microscopic periodic and crystal symmetry of a crystal, a larger volume primitive cell, called "crystal cell". Lattice points may locate corner, centers of the body and face. Each cell contains one or more lattice point on average, and its basis vector is written as $\vec{a},\vec{b},\vec{c}$

维格纳-赛兹原胞Wigner Seitz primitive cell

是以一格点为中心，画中心到最近邻格点的连线的垂直平分面，围成的多面体称维格纳-赛兹原胞。它也是体积最小的原胞，其格点只在中心，每个原胞平均只含一个格点；A polyhedron formed by perpendicular bisectors of lines connecting the center of a lattice point to its nearest neighbor is called a Wigner Seitz primitive cell. The smallest primitive cell in terms of volume, with its lattice points located only in the center, and each primitive cell contains an average of only one lattice point;

体心立方结构的各种原胞/晶胞



体心立方晶胞

BCC unit cell

体心立方原胞

BCC primitive cell

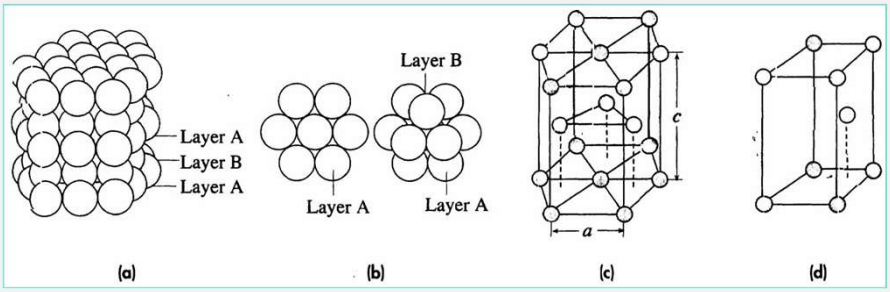
体心立方维格纳-赛兹原胞

BCC Wigner Seitz primitive cell

常见晶体结构：HCP

Hexagonal close-packed structure

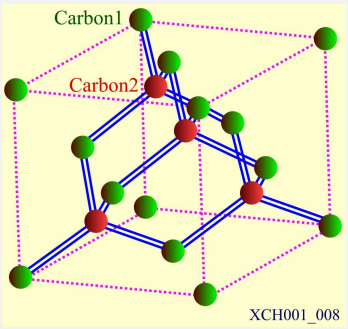
六方密堆积结构



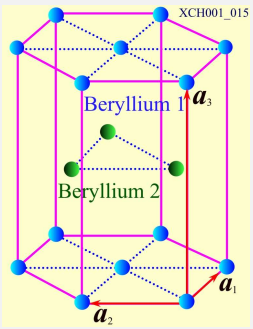
复式晶格 Complex lattice

a) 同种原子的相同晶格结构相互错位镶嵌而成

金刚石结构的C、Si、Ge

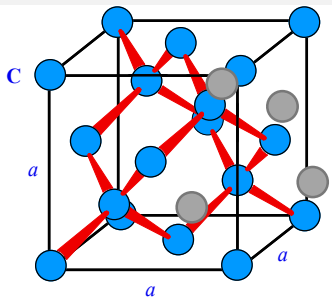


六角密排结构Be、Mg、Zn



b) 不同原子的相同晶格结构相互错位镶嵌而成

a）同种原子复式晶格的例子（面心立方子晶格）：
Diamond cubic crystal structure 金刚石立方晶体结构



Diamond structure is composed of two FCC sublattices from the same atom that are displaced by 1/4 of their length along the diagonal of the cubic space

同种原子的两个面心立方沿着对角线位移1/4对角线长度

The diamond unit cell is cubic. The cell has eight atoms. Grey Sn (α-Sn) and the elemental semiconductors Ge and Si have this crystal structure.

Diamond cubic crystal structure金刚石立方晶体结构. Eight atoms at the corner, six atoms at the center of each face, four atoms in the cell.

a= $\frac{8r}{\sqrt{3}}$; Cn=4

Silicon: same atoms in a diamond lattice

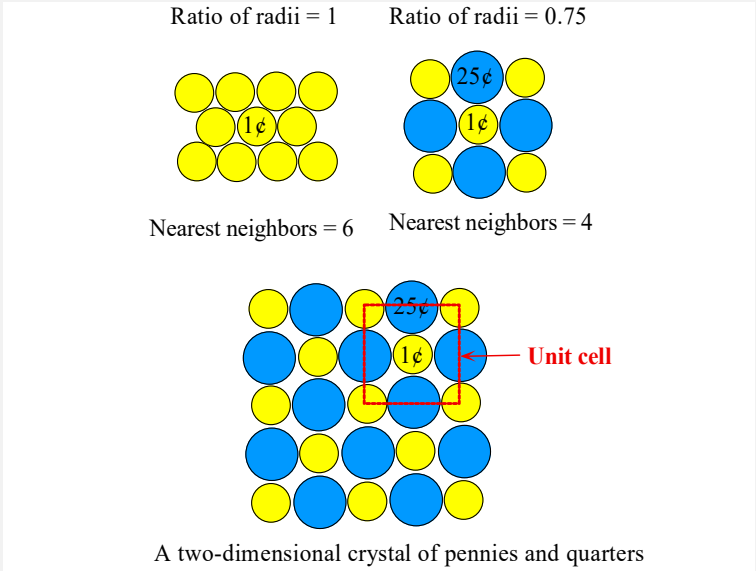


A silicon ingot is a single crystal of Si.





b）不同原子构成复合晶格情况：Ionic solids 离子型晶体

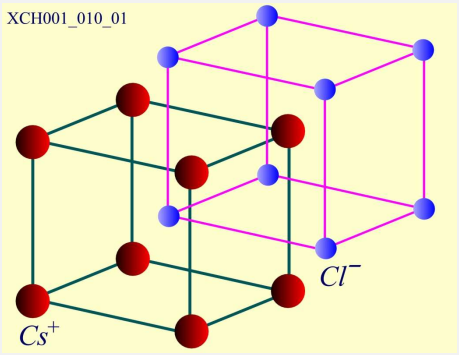


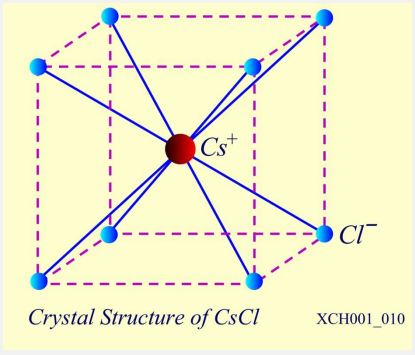
A two-dimensional crystal of pennies and quarters

Packing of coins on a table top to build a two dimensional crystal.

***CsCl*的复式晶格 complex lattice**

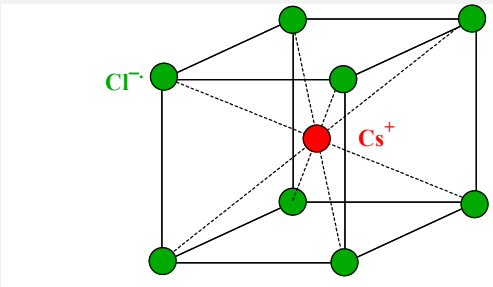
—— ***CsCl*结构是由两个简立方的子晶格彼此沿立方体空间对角线位移1 / 2 的长度套构而成**

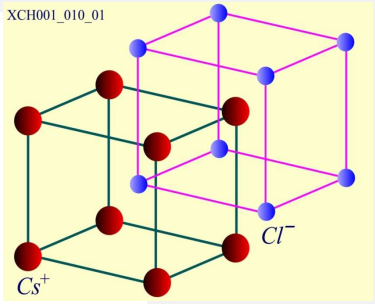




The CsCl structure is composed of two simple cubic sublattices that are displaced by 1/2 of their length along the diagonal of the cubic space

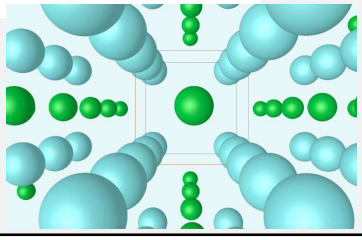
CsCl structure :CsCl 结构





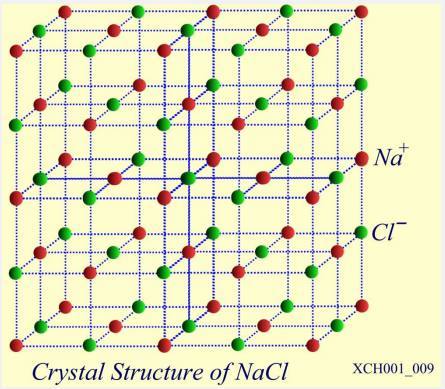
A possible reduced sphere unit cell for the CsCl crystal. An alternative unit cell may have Cs^+ and Cl^- interchanged. Examples: CsCl, CsBr, CsI, TlCl, TlBr, TlI.

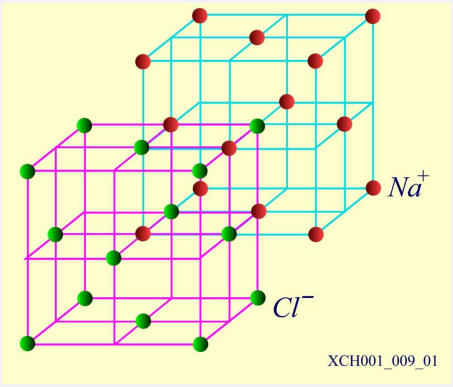
Cn=8



复式格子的特点：不同等价原子各自构成相同的简单晶格（子晶格），复式格子由它们的子晶格相套而成

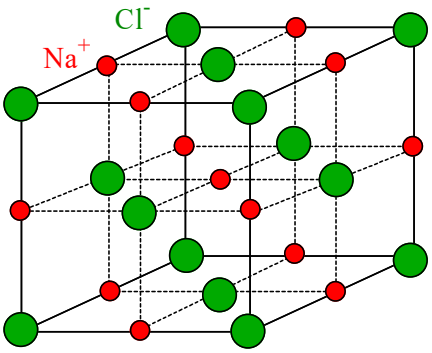
NaCl晶格 —— Na^+和Cl^-两个面心立方子晶格

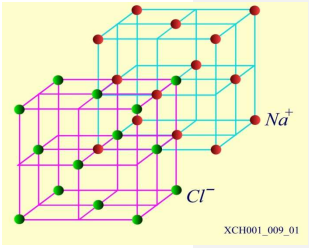




two FCC sublattices； shifted half of lattice constant

NaCl unit cel l氯化钠晶胞



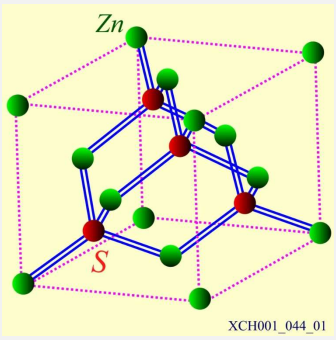


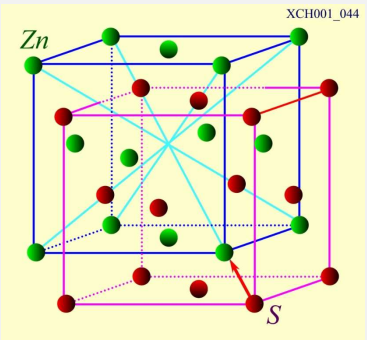
A possible reduced sphere unit cell for the NaCl (rock salt) crystal.
An alternative unit cell may have Na^+ and Cl - interchanged.
Examples: AgCl, CaO, CsF, LiF, LiCl, NaF, NaCl, KF, KCl, MgO

Cn=6

ZnS的复式晶格 ZnS complex lattice

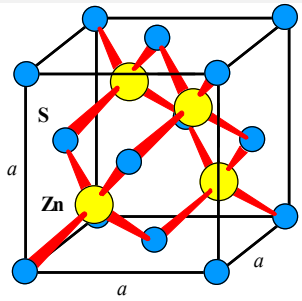
立方系的ZnS —— S和Zn两个面心立方结构的子晶格沿空间对角线位移 1 / 4 的长度套构而成

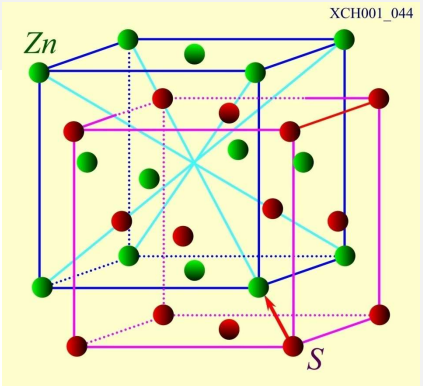




The ZnS structure is composed of two FCC sublattices of Zn and S that are displaced by 1/4 of their length along the diagonal of the cubic space (equal to: two sublattices of different atoms in a diamond structure)

Zinc blende structure闪锌矿结构

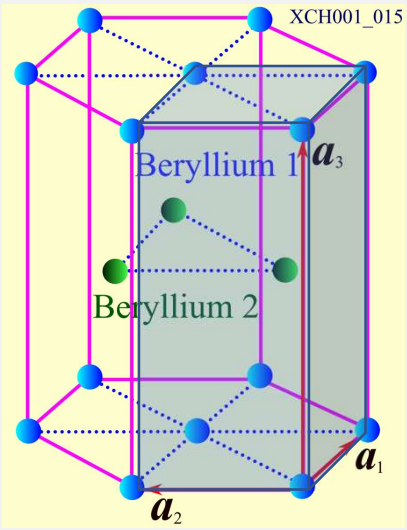




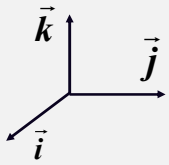
The Zinc blende (ZnS) cubic crystal structure. Many important compound crystals have the zinc blende structure. Examples: AlAs, GaAs, GaP, GaSb, InAs, InP, InSb, ZnS, ZnTe.

问：同种原子的情况下称之为？ 金刚石结构

Hexagonal close-packed structure(HCP)六方密堆积结构



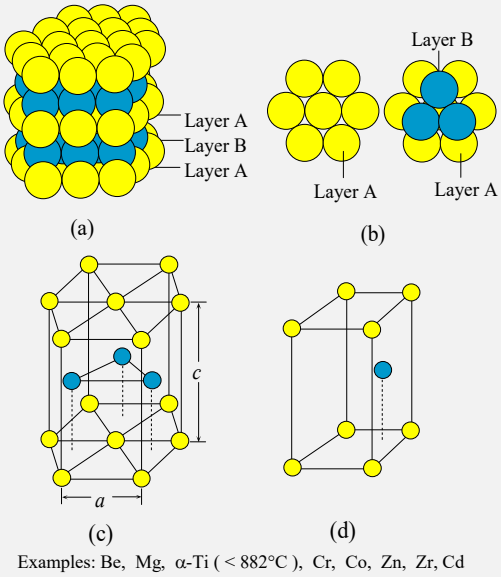
Primitive cell and base vector



Base vectors: $\vec{a}_1, \vec{a}_2, \vec{a}_3$

$$a_1 = a_2 \neq a_3$$

Hexagonal close-packed structure(HCP)六方密堆积结构



Examples: Be, Mg, α-Ti (< 882°C), Cr, Co, Zn, Zr, Cd

The Hexagonal Close Packed (HCP) Crystal Structure.

(a) The Hexagonal Close Packed (HCP) Structure. A collection of many Zn atoms. Color difference distinguishes layers (stacks).

(b) The stacking sequence of closely packed layers is ABAB

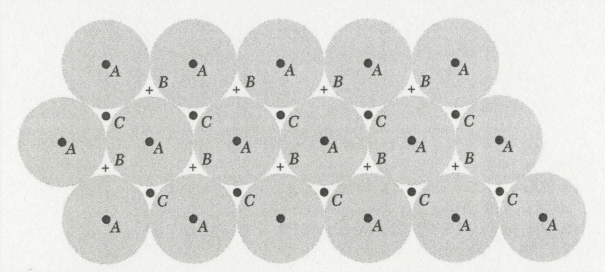
(c) A unit cell with reduced spheres

(d) The smallest unit cell with reduced spheres.
Cn=12

Hexagonal close-packed structure(HCP)六方密堆积结构,The atomic packing density in the HCP crystal structure is 74 percent, same as FCC structure.

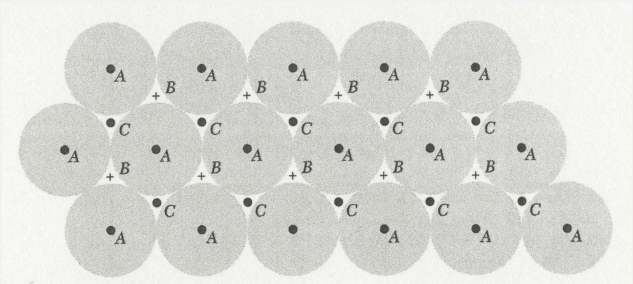
Hexagonal close-packed structure(HCP)六方密堆积结构

把原子想象成刚性的硬球，让这些硬球在一个平面上按最紧密的排列方式排列起来，那么只可能是这种排列方式。这种排列方式形成的面，称为密排面。Imagine atoms as rigid hard spheres arranged in the closest possible arrangement on a plane, and this is the only possible arrangement. The surface formed by this arrangement is called a close packed surface.



把这些密排面按最紧密的方式堆积起来，即构成密堆积结构。Stacking these densely packed surfaces in the most compact way forms a dense stacking structure.

堆积方式有两种 two stacking styles:

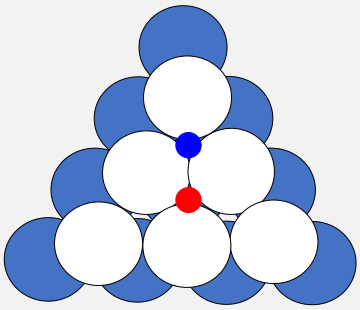


1) 六角密堆积 (hcp)结构排列方式为 ABABAB……。

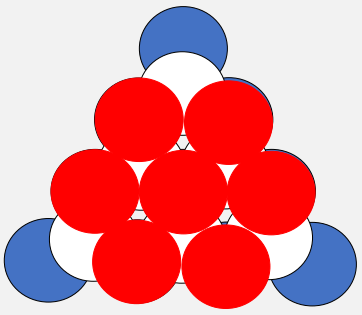
2) 面心立方（fcc）结构排列方式为 ABCABCABC……。

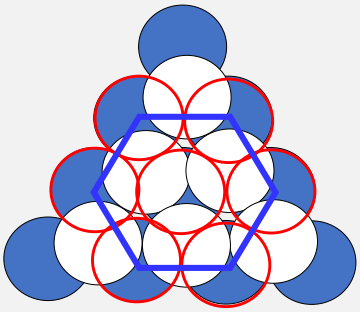
对于这两种结构，其总体积中被球占据的体积比率都是0.74。除此之外无论是规则还是不规则的堆积，都不可能得到比fcc和hcp更密的堆积。 For both structures, the volume ratio occupied by balls in their total volume is 0.74. Besides, whether it is regular or irregular stacking, it is impossible to obtain a denser stacking than fcc and hcp

Hexagonal close-packed structure(HCP)六方密堆积结构



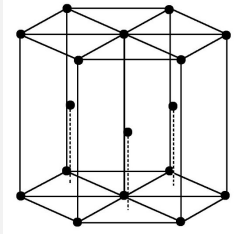
第三层有两种排列方式
on the third layer, there are two possible stacking positions

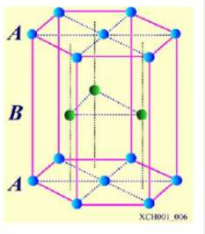


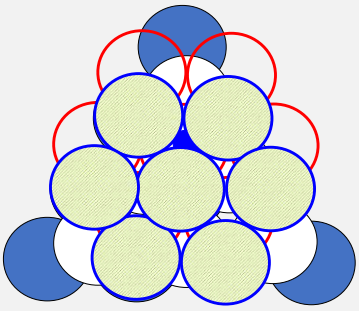


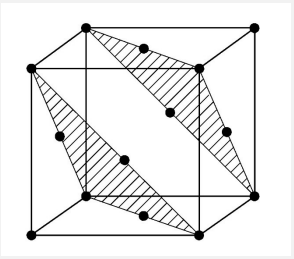
六角密排hcp
ABABAB……

配位数Cn=12









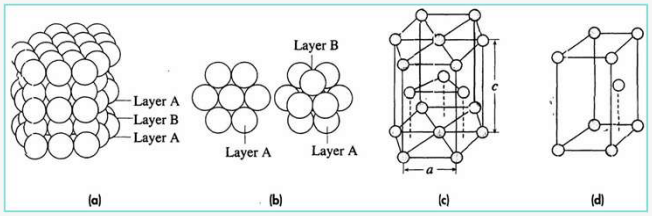
面心立方fcc(立方密排)
ABCABC····

配位数Cn=12

Table 1.3 Properties of some important crystal structures

Crystal Structure	a and R (R is the Radius of the Atom)	Coordination Number (CN)	Number of Atoms per Unit Cell	Atomic Packing Factor	Examples
Simple cubic	$a = 2R$	6	1	0.52	No metals (Except Po)
BCC	$a = \frac{4R}{\sqrt{3}}$	8	2	0.68	Many metals: α–Fe, Cr, Mo, W
FCC	$a = \frac{4R}{\sqrt{2}}$	12	4	0.74	Many metals: Ag, Au, Cu, Pt
HCP	$a = 2R$ $c = 1.633a$	12	2	0.74	Many metals: Co, Mg, Ti, Zn
Diamond	$a = \frac{8R}{\sqrt{3}}$	4	8	0.34	Covalent solids: Diamond, Ge, Si, α-Sn
Zinc blende		4	8	0.34	Many covalent and ionic solids. Many compound semiconductors. ZnS, GaAs, GaSb, InAs, InSb
NaCl		6	4 cations	0.67	Ionic solids such as NaCl, AgCl, LiF, MgO, CaO
			4 anions	(NaCl)	Ionic packing factor depends on relative sizes of ions.
CsCl		8	1 cation 1 anion		Ionic solids such as CsCl, CsBr, CsI

同种原子的 HCP结构



层状堆叠模式

原子以六边形层紧密排列，遵循**ABAB…**的重复堆叠序列，形成高度对称的六方晶格结构，无面心立方的立方对称性。

极致空间利用率

属于密排结构，原子占据了晶胞**74%**的体积，剩余26%为空隙。这种紧密堆积赋予了金属良好的延展性和高密度特性。

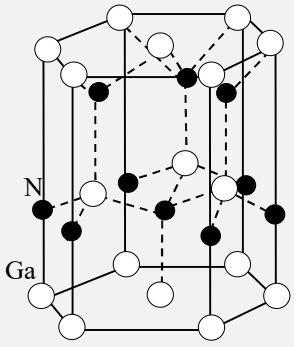
工业应用实例

典型代表包括锌(Zn)、镁(Mg)、钛(Ti)及镉(Cd)。其中钛合金因兼具高强度与低密度，是航空航天工程的关键材料。

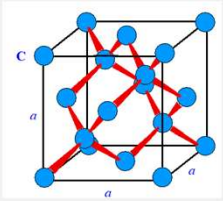
不同种原子的HCP结构

氮化物半导体 III-nitride semiconductors

纤锌矿结构 wurtzite structure，六角晶系(HCP)，由六角排列的双原子(two different atoms)层堆叠而成

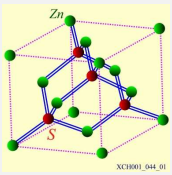


金刚石结构（同种原子）
(diamond structure)



金刚石/Si/Ge…..

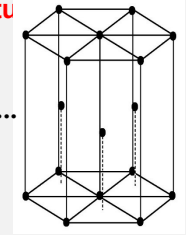
闪锌矿结构（不同种原子）
(Zinc blend structure)



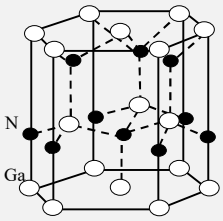
ZnS/GaAs/InP…….

六方密堆积结构 (同种原子)
Hexagonal close-packed structu

石墨/Cr/Zn/Mg…..

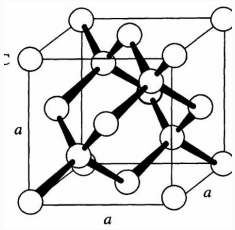


纤锌矿结构 （不同种原子）
wurtzite structure



GaN/AlN…..

常见晶体结构：陶瓷与半导体



01 金刚石立方结构

由两个全同原子基元构成的面心立方（FCC）晶格，是硅（Si）、锗（Ge）等元素半导体的典型晶体结构，奠定了现代微电子的基础。

8个原子/晶胞

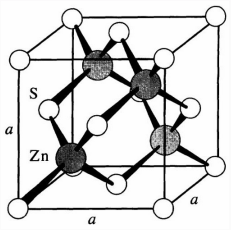
顶点、面心及内部位置的原子共同构成，空间利用率高。

强共价键合

原子间通过共用电子对结合，键能高，赋予材料高硬度与熔点。

四面体配位

每个原子被4个等距原子包围，形成稳定的正四面体空间构型。



02 闪锌矿 (ZnS) 结构

由两种不同原子基元构成的FCC晶格，是砷化镓（GaAs）、磷化铟（InP）等化合物半导体的核心结构，广泛应用于光电子领域。

4+4 原子配比

晶胞共含8个原子，两类原子各占4个，满足电中性平衡。

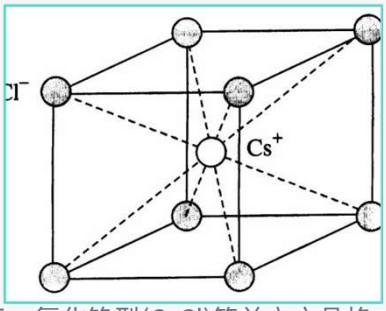
混合键合特性

兼具共价键的方向性与离子键的静电作用，载流子迁移率优异。

极性晶格特征

原子电负性差异导致晶格具有极性，深刻影响材料的表面能与界面态。

常见晶体结构：陶瓷 - 岩盐



图示：氯化铯型(CsCl)简单立方晶格结构模型
展示了体心阳离子与顶点阴离子的空间排布

01. 岩盐 (NaCl) | Rock Salt Structure
由两个互相穿插的面心立方(FCC)晶格构成，分别承载阳离子与阴离子。具有典型的**离子键合**特征，阴阳离子配位数均为6，结构对称性高，是陶瓷材料中最基础的晶体构型之一。

02. 氯化铯 (CsCl) | Cesium Chloride Structure
属于简单立方晶格，晶胞基元由1个体心阳离子和1个顶点阴离子组成。其配位数为8，区别于岩盐结构，这种构型常见于半径较大的阳离子与小半径阴离子的组合，具有更高的堆积密度。

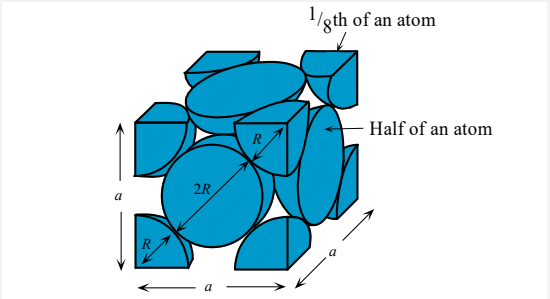
例题 example 1.13
Consider the FCC unit cell of the copper crystal shown in the right.

a. How many atoms are there per unit cell?

b. If r is the radius of the Cu atom, show that the lattice parameter a is given by $a=r2\sqrt{2}$.

c. Calculate the APF.

d. Calculate the atomic concentration(number of atoms per unit volume) in Cu and the density of the crystal given that the atomic mass of Cu is 63.55g mol^{-1} and the radius of the Cu atom is 0.128nm



The FCC unit cell. The atomic radius is *R* and the lattice parameter is *a*

Atomic packing factor原子堆积(填充）因子, APF=$\frac{Volume\ of\ atomes\ in\ unit\ cell}{volume\ of\ unit\ cell}$

SOLUTION

a. There are four atoms per unit cell. The Cu atom at each corner is shared with eight other adjoining unit cells. Each Cu atom at the face center is shared with the neighboring unit cell. Thus, the number of atoms in the unit cell = 8 corners ($\frac{1}{8}$ atom) + 6 faces ($\frac{1}{2}$ atoms) = 4 atoms.

b. Consider the unit cell shown in Figure 1.38 and one of the cubic faces. The face is a square of side a and the diagonal is $\sqrt{a^2 + a^2}$ or $a\sqrt{2}$. The diagonal has one atom at the center of diameter $2R$, which touches two atoms centered at the corners. The diagonal, going from corner to corner, is therefore $R + 2R + R$. Thus, $4R = a\sqrt{2}$ and $a = 4R/\sqrt{2} = R2\sqrt{2}$. Therefore, $a = 0.3620$ nm.

c. $$\text{APF} = \frac{(\text{Number of atoms in unit cell}) \times (\text{Volume of atom})}{\text{Volume of unit cell}}$$

$$= \frac{4 \times \frac{4}{3}\pi R^3}{a^3} = \frac{\frac{4^2}{3}\pi R^3}{(R2\sqrt{2})^3} = \frac{4^2\pi}{3(2\sqrt{2})^3} = 0.74$$

d. In general, if there are x atoms in the unit cell, the atomic concentration is

$$n_{at} = \frac{\text{Number of atoms in unit cell}}{\text{Volume of unit cell}} = \frac{x}{a^3}$$

Thus, for Cu

$$n_{at} = \frac{4}{(0.3620 \times 10^{-7}\ \text{cm})^3} = 8.43 \times 10^{22}\ \text{cm}^{-3}$$

There are x atoms in the unit cell, and each atom has a mass of M_{at}/N_A grams. The density ρ is

$$\rho = \frac{\text{Mass of all atoms in unit cell}}{\text{Volume of unit cell}} = \frac{x\left(\frac{M_{at}}{N_A}\right)}{a^3}$$

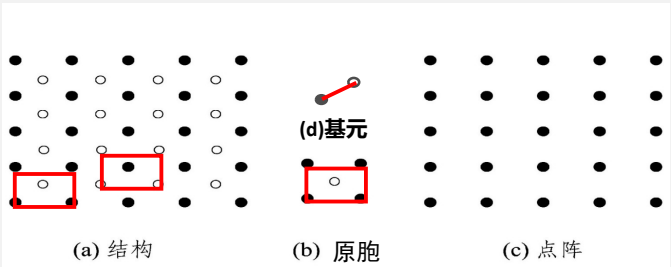
that is, $$\rho = \frac{n_{at}M_{at}}{N_A} = \frac{(8.43 \times 10^{22}\ \text{cm}^{-3})(63.55\ \text{g mol}^{-1})}{6.022 \times 10^{23}\ \text{mol}^{-1}} = 8.9\ \text{g cm}^{-3}$$

The expression $\rho = (n_{at}M_{at})/N_A$ is independent of the crystal structure.

（四）布拉伐（布拉维）格子（Bravais lattice)

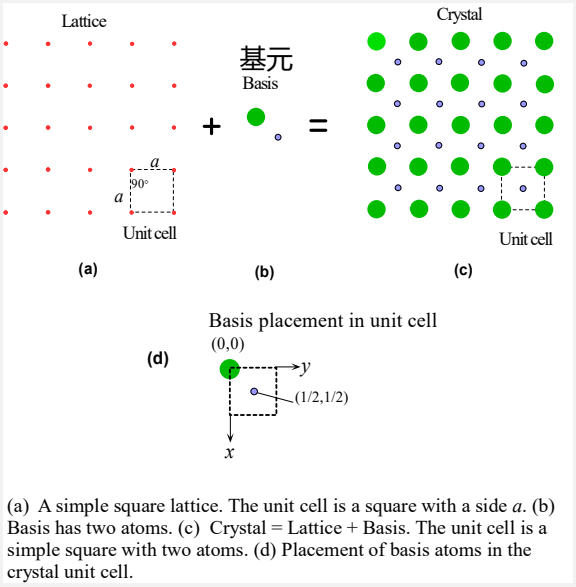
晶体可以看作是在布拉维格子（Lattice）的每一个格上放上一组原子（Basis 基元）构成的

晶体结构=点阵+基元
Crystal =Lattice +Basis



Bravais lattices 布拉维格子

空间点阵



(a) A simple square lattice. The unit cell is a square with a side *a*. (b) Basis has two atoms. (c) Crystal = Lattice + Basis. The unit cell is a simple square with two atoms. (d) Placement of basis atoms in the crystal unit cell.

A geometric point(格点） --> A representative point of a Basis(基元)

Lattice：An infinite periodic array of geometric points in space defines a space lattice（空间点阵，布拉维格子).

Crystal = Lattice +Basis

a: lattice parameter, 晶胞参数

晶体方向（米勒指数）

01 目的 | Purpose
在晶体学中，米勒指数用于唯一地指定晶体点阵中的任意方向，是描述晶体结构与材料物理性质的基础语言。

01确定原点
选择晶格点作为坐标系原点，通常选在晶胞的顶点，确保计算的基准清晰。

02读取坐标
找出方向直线上任意一点的坐标值(x, y, z)，该值需表示为晶格参数(a, b, c)的整数倍。

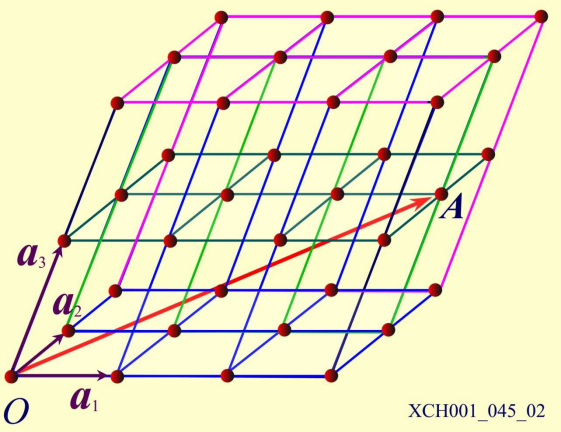
03约简整数
将坐标化为互质的最小整数集(u, v, w)，消除公约数，得到最简整数比。

04规范表示
将约简后的整数用方括号括起，即 [uvw]，这是晶体方向米勒指数的标准书写形式。

实例演示 | Example
若有一直线通过点 (1a, 2b, 1c)，该坐标已是最简整数比，无需约分。因此，该晶体方向的米勒指数直接表示为：**[121]**

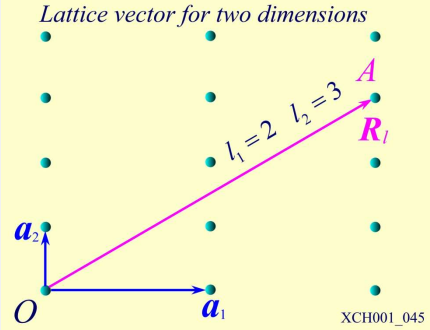
用下面公式可以表示所有格点 all lattice points can be expressed by the following:

$$\vec{R}_l = l_1\vec{a}_1 + l_2\vec{a}_2 + l_3\vec{a}_3$$



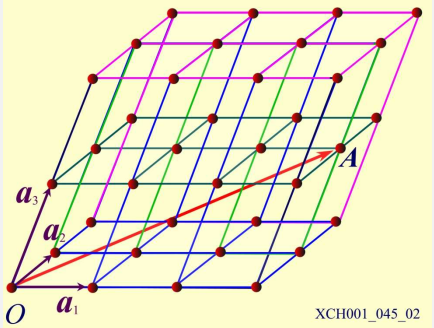
简单晶格，任一原子A的位矢 $\vec{R}_l = l_1\vec{a}_1 + l_2\vec{a}_2 + l_3\vec{a}_3$

$\vec{R}_l = 2\vec{a}_1 + 3\vec{a}_2$



Two dimensional

$\vec{R}_l = 3\vec{a}_1 + \vec{a}_2 + \vec{a}_3$



Three dimensional

复式晶格 complex lattices：任一原子A的位矢

$$\vec{R}_l = \vec{r}_a + l_1\vec{a}_1 + l_2\vec{a}_2 + l_3\vec{a}_3, \quad \alpha = 1,\ 2,\ 3 \ldots\ldots$$

$\vec{r}_\alpha$ 原胞中各种等价原子之间的相对位移 relative displacement

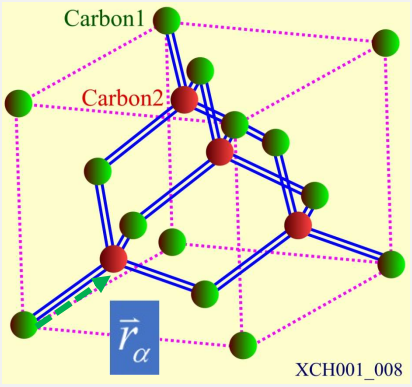
—— 金刚石晶格

—— 碳1位置 $l_1\vec{a}_1 + l_2\vec{a}_2 + l_3\vec{a}_3$

—— 碳2位置

$$\vec{\tau} + l_1\vec{a}_1 + l_2\vec{a}_2 + l_3\vec{a}_3$$

对角线位移 $|\vec{\tau}| = 1/4$



B

晶体平面（米勒指数）

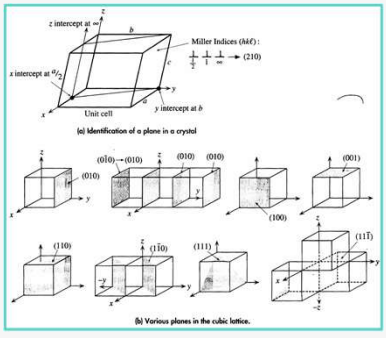
01 核心目的 | Purpose

用于唯一且标准化地指定晶体结构中的任意原子平面，是材料科学中描述晶体取向、分析衍射图谱的基础语言。

1. **确定截距：**找出平面与 x, y, z 轴的交点，结果以晶格常数 (a, b, c) 的倍数表示。
2. **取其倒数：**将各轴上的截距值取倒数，若平面与某轴平行，则截距为 ∞，其倒数为 0。
3. **化简整数：**消去分数并化为最简整数比，得到一组互质的整数 (h, k, ℓ)。

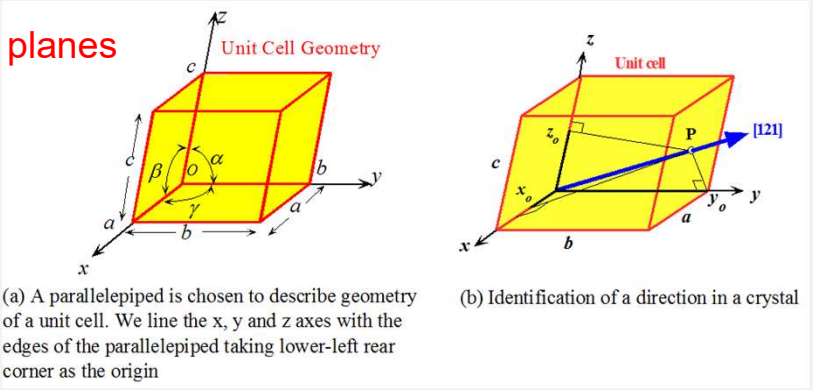
实例解析：(210) 晶面

某平面与 x 轴交于 ½a，与 y 轴交于 1b，与 z 轴平行（截距 ∞）。其截距为 (½, 1, ∞)，取倒数得 (2, 1, 0)，即该平面为**(210) 晶面**。



图示展示了立方晶格中不同晶面的几何形态与米勒指数的对应关系，直观体现了指数如何反映平面在晶体坐标系中的位置与取向。

Crystal directions and planes
晶向与晶面



(a) A parallelepiped is chosen to describe geometry of a unit cell. We line the x, y and z axes with the edges of the parallelepiped taking lower-left rear corner as the origin

(b) Identification of a direction in a crystal

$l_1\vec{a} + l_2\vec{b} + l_3\vec{c}$

Geometry of the unit cell (晶胞参数）=side（边长）a,b,c + angle（角度）σ,β,γ

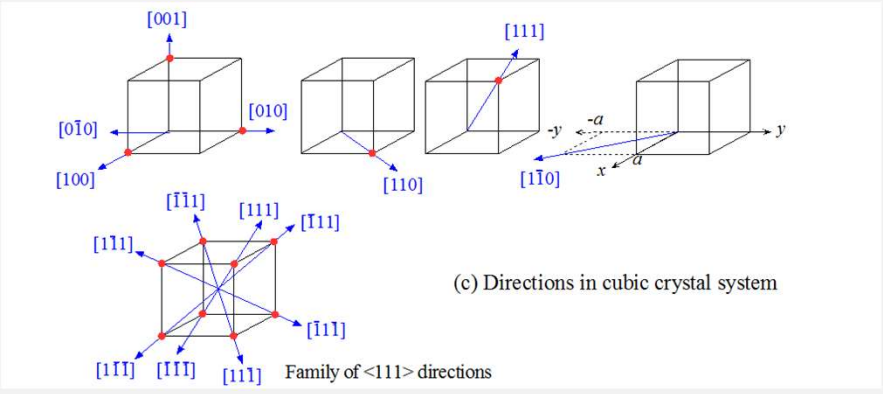
For example:

$x_0,y_0,z_0: \frac{1}{2}a,\ b,\ \frac{1}{2}c$

P is at:

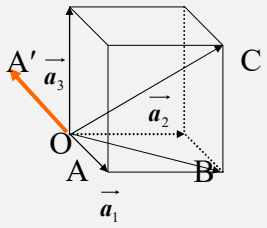
$x_p,y_p,z_p: \frac{1}{2},\ 1,\ \frac{1}{2}$

约分为最小整数即是晶向指数：[uvw] -> [121] 方括号



(c) Directions in cubic crystal system

Family of <111> directions



立方晶格中的晶向

OA------[100] [$\bar{1}$00]

OB------[110] [$\bar{1}\bar{1}$0]

OC------[111] [$\bar{1}\bar{1}\bar{1}$]

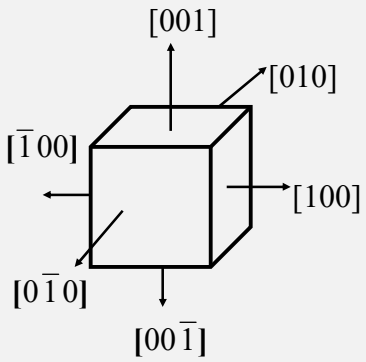
负指数在数字上加一横线。

A family of directions（晶向族）

Certain directions in the crystal are equivalent. For example, along the [010]and [001], all directional properties of a material are same. All of these directions constitute a family of directions（晶向族）。

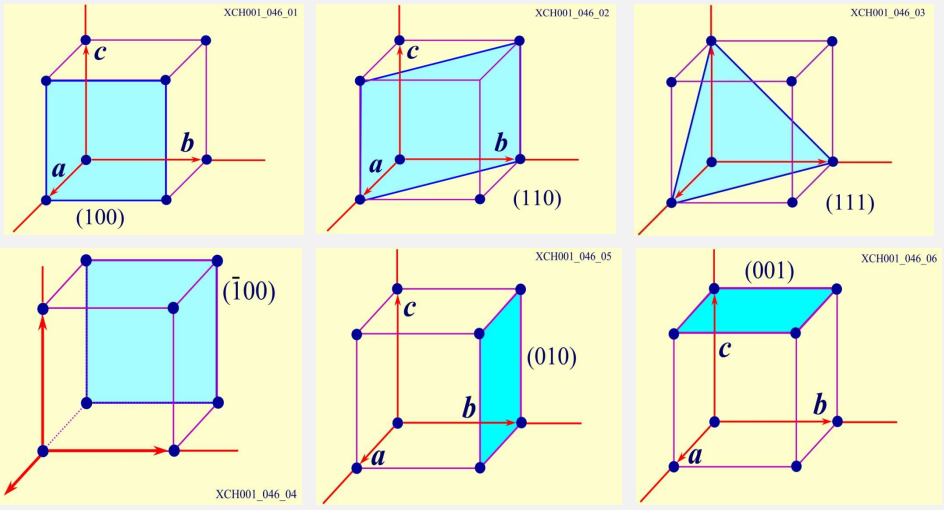
考虑到晶格的对称性，某些晶向可以只是方向不同，而材料性质是相同的，这类晶向称为等效晶向。等效晶向的全体用< >表示。

如立方晶格中的[$\bar{1}$00], [0$\bar{1}$0], [00$\bar{1}$],[100], [010],[001]六个晶向可用符号〈 100 〉表示其晶向族，它们是等效晶向。三角符号



[100]及等效晶向

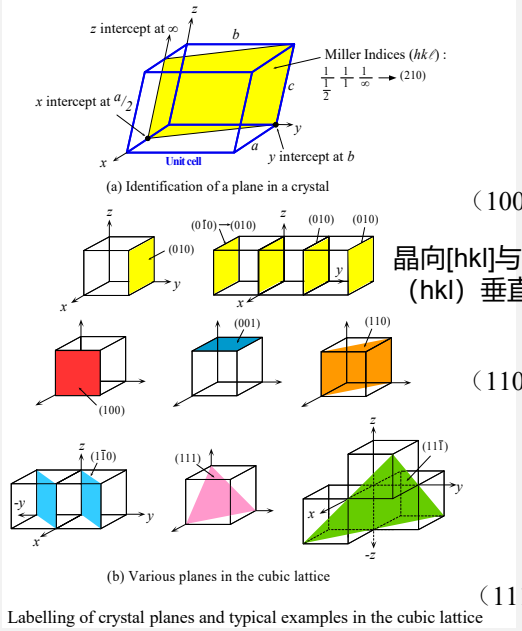
planes 晶面, Miller indices of a plane 密勒指数

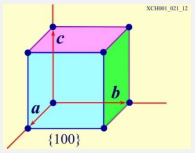


密勒指数表述晶面，晶面在xyz轴的截距取倒数，再化简为整数。

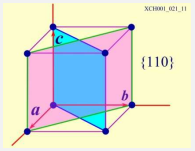
1/2a, 2b, œ -----> 2, ½ , 0 -----> 4,1, 0

planes 晶面, Miller indices of a plane 密勒指数

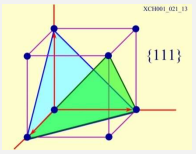




（100）面等效的晶面数分别为：6个，表示为：{100}



（110）面等效的晶面数分别为：6个，表示为：{110}



（111）面等效的晶面数分别为：4个，表示为：{111}

{ } family of planes, 晶面族

米勒指数 (Miller indices), 晶面（族），晶向（族）

几个参考书 a few references

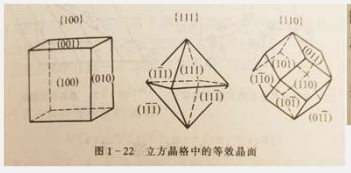
立方晶系中立方体的 6 个外表面的晶面指数分别为(100)、(010)、(001)、($\bar{1}$00)、(0 $\bar{1}$0)、(00 $\bar{1}$)。由于对称性，这些晶面是等效的，它们的面间距和晶面上原子的分布完全相同。在许多晶系中都有由对称性联系起来的等效晶面族，这些等效晶面族用{*hkl*}表示，例如上面所说的等效晶面表示成{100}。

{100}晶面族有6个面

等な六つの面の組を表わすのに {100} のように中かっこを用いて表わす。一

同一晶体中面间距相同的晶面族，由于在垂直于晶面的方向上，其宏观性质相同，所以常称它们为同族晶面族，并常以大括号表示之。如立方晶系晶体的晶面族{111}包括(111)、($\bar{1}$ $\bar{1}$ $\bar{1}$)、($\bar{1}$11)、(1 $\bar{1}$1)、(11 $\bar{1}$)、($\bar{1}$ $\bar{1}$1)、(1 $\bar{1}$ $\bar{1}$)和($\bar{1}$1 $\bar{1}$)。

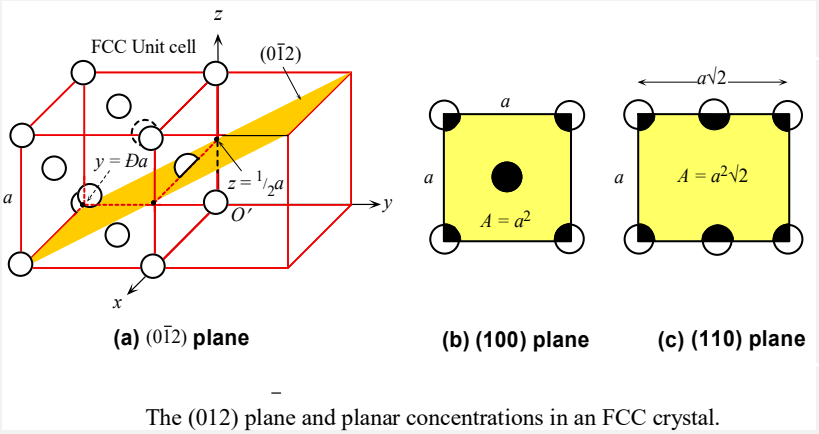
{111}晶面族有8个面



因为符号相反的晶面指数所标志的晶面是相互平行的，所以对标志晶格里面的晶面来讲，是没有什么区别的，因而|100|、|110|、|111|的等效晶面数分别为 3、4、6。符号相反的晶面指数只是在区别晶体的外表面时才是有意义的。

{100}{111}{110}晶面族分别有3，4，6个面

Miller indices and planar concentration 密勒指数与晶面浓度



The (012) plane and planar concentrations in an FCC crystal.

Calculate the planar concentration on (100), (110).

$$n_{(100)} = \frac{4*\frac{1}{4}+1}{a^2} = \frac{2}{a^2} = \frac{2}{(0.3620*10^{-9}m)^2} = 15.3\text{原子.nm}^{-2}$$

$$n_{(110)} = \frac{4*\frac{1}{4}+2*\frac{1}{2}}{a*(\sqrt{2}a)} = \frac{2}{\sqrt{2}a^2} = \frac{2}{\sqrt{2}(0.3620*10^{-9}m)^2} = 10.8\text{原子.nm}^{-2}$$

基本晶体结构

密勒指数

- ($\bar{h}kl$)：代表在x轴上截距为负的平面，如 ($\bar{1}$00)
- {hkl}：代表相对称的平面群，如在立方对称平面中，可用{100}表示(100)，(010)，(001)，($\bar{1}$00)，(0$\bar{1}$0)，(00$\bar{1}$)六个平面。
- [hkl]：代表一晶体的方向，如[100]方向定义为垂直于(100)平面的方向，即表示x轴方向。而[111]则表示垂直于(111)平面的方向。
- <hkl>：代表等效方向的所有方向组，如<100>代表[100]、[010]、[001]、[$\bar{1}$00]、[0$\bar{1}$0]、[00$\bar{1}$]六个等效方向的族群。

例　如图所示平面在沿着三个坐标轴的方向有三个截距a、3a、2a，其的倒数分别为1/a、1/3a和1/2a。它们的最简单整数比为6:2:3（每个分数乘6a所得）。因此这个平面可以表示为(623)平面。

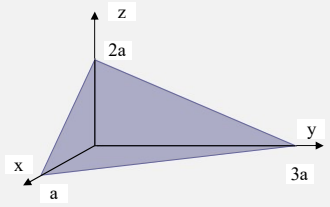


表 1.4　碳的同素异形体(ρ是密度,Y是弹性模量或杨氏模量)

	石墨	金刚石	富勒烯晶体
结构	层内共价键,层间范德瓦耳斯力;六方晶胞	共价键网络。金刚石晶体结构	共价键结合的 C_{60} 球状分子通过范德瓦耳斯力构成 FCC 结构
电学和热学性质	良好的导电性,热导率可同金属相比	非常好的电绝缘性,优异的热导率,是银或铜的 5 倍以上	半导体。同碱金属形成的化合物(如 K_3C_{60})具有超导性
机械性质	润滑剂。可机械加工石墨块体:$Y\approx27$GPa,$\rho=2.25\ g\cdot cm^{-3}$	最硬的材料。$Y\approx827$ GPa $\rho=3.25\ g\cdot cm^{-3}$	软材料。$Y\approx18$ GPa $\rho=1.65\ g\cdot cm^{-3}$
评述	大气压力下稳定的同素异形体	高压同素异形体	实验室合成的,产生于不完全燃烧的煤灰中
用途或潜在的用途	冶金坩埚,焊接电极,加热体,电接触,耐火应用	切削工具应用:金刚石砧,金刚石薄膜包覆钻头、刀片、轴承等;珠宝;可能用于薄膜半导体器件,比如电荷载流子迁移率大	将来可能在半导体或超导体方面有应用

Table 1.4 Crystalline allotropes of carbon (ρ is the density and Y is the elastic modulus or Young's modulus)

	Graphite	Diamond	Buckminsterfullerene Crystal
Structure	Covalent bonding within layers. Van der Waals bonding between layers. Hexagonal unit cell.	Covalently bonded network. Diamond crystal structure.	Covalently bonded C_{60} spheroidal molecules held in an FCC crystal structure by van der Waals bonding.
Electrical and thermal properties	Good electrical conductor. Thermal conductivity comparable to metals.	Very good electrical insulator. Excellent thermal conductor, about five times more than silver or copper.	Semiconductor. Compounds with alkali metals (*e.g.*, K_3C_{60}) exhibit superconductivity.
Mechanical properties	Lubricating agent. Machinable. Bulk graphite: $Y \approx 27$ GPa $\rho = 2.25\ g\ cm^{-3}$	The hardest material. $Y = 827$ GPa $\rho = 3.25\ g\ cm^{-3}$	Mechanically soft. $Y \approx 18$ GPa $\rho = 1.65\ g\ cm^{-3}$
Comment	Stable allotrope at atmospheric pressure	High-pressure allotrope.	Laboratory synthesized. Occurs in the soot of partial combustion.
Uses, potential uses	Metallurgical crucibles, welding electrodes, heating elements, electrical contacts, refractory applications.	Cutting tool applications. Diamond anvils. Diamond film coated drills, blades, bearings, etc. Jewelry. Heat conductor for ICs. Possible thin-film semiconductor devices, as the charge carrier mobilities are large.	Possible future semiconductor or superconductivity applications.

同素异形体与碳单质 Allotropy and the three phases of carbon

01 定义 | Definition

指一种元素或化合物能以两种或多种不同晶体结构存在的现象，这些变体在不同温度或压力下稳定性不同，从而赋予材料截然不同的物理与化学性能。

The ability of a material to exist in more than one crystal structure (polymorph), which results in distinct physical and chemical properties under varying conditions.

α-Fe (BCC)

稳定于**912 °C 以下**。具有体心立方结构，是铁磁性的来源，也是钢材在常温环境下的主要组成相，强度适中且具有良好的磁性。

γ-Fe (FCC)

稳定于**912 ~ 1400 °C**。具有面心立方结构，原子密排度更高，因此塑性极佳，是奥氏体不锈钢和高温合金的关键组成相。

δ-Fe (BCC)

稳定于**1400 °C 以上**直至熔点(1538°C)。再次转变为体心立方结构，是钢铁冶炼和连铸过程中高温区的重要相组成。

同素异形体：碳(同种物质不同结构)

金刚石 | Diamond

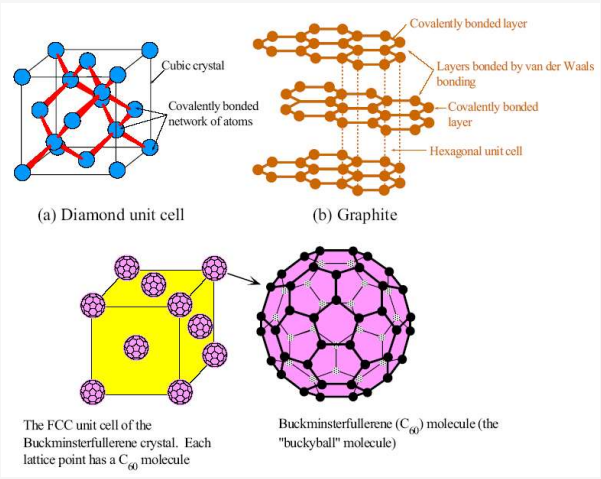
具有坚固的四面体共价键网状结构，是自然界天然存在的最硬物质，化学性质稳定，同时也是极佳的绝缘体。

石墨 | Graphite

由平面六边形碳原子层堆叠而成，层间以微弱的范德华力结合，质软滑腻，且因自由电子存在而具有良好导电性。

富勒烯 (C_{60}) | Fullerene

由60个碳原子构成的空心球状分子，形似足球，兼具无机分子与纳米材料的特性，在超导、催化等领域潜力巨大。



立方金刚石以外，还有六方金刚石。六方金刚石晶体其晶胞参数为a=251pm,c=412pm。在这种金刚石中，碳原子的成键方式和C—C键键长均和立方金刚石相似。立方金刚石采用交叉式排列，在C—C 键中具有对称中心对称性；六方金刚石中一部分C—C键采用重叠式构象，平行c轴的C—C键中心点具有镜面对称性。

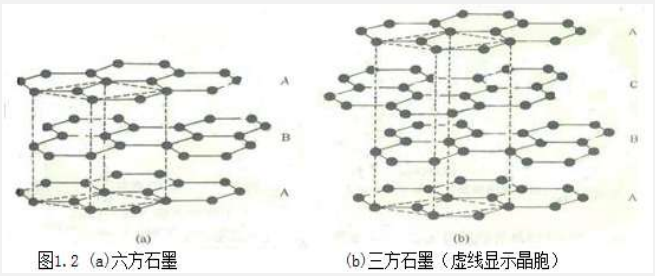
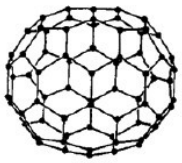


图1.2 (a)六方石墨　　(b)三方石墨（虚线显示晶胞）



球碳C_{70}　fullerene-C_{70}　由70个C原子组成椭球形的分子。可用石墨放电气化凝聚的黑烟或用苯不完全燃烧形成的黑烟，经萃取结晶所得的C_{60}和C_{70}混合物，进一步用结晶法或液相色谱法，或其他方法分离制得。晶体呈棕黑色，溶于有机溶剂。分子中C原子组成12个五元环面、25个六元环面、70个顶点和105条棱边的椭球形多面体属D_{5h}点群，见图。许多含C_{70}基团的化合物已经制得。[P]

铁： Polymorphism多形性 or allotropy同素异形: have more than one crystal structure.
<912°C;　BCC
912<->1400°C;　FCC
>1400;　BCC

Transition temperature相变温度: one allotrope changes to another at this temperature.